



Feathered wing

Fossilized skeleton of *Archaeopteryx*

1861

Dinosaur to bird

The almost complete fossil of a creature with wings, feathers, and a toothed beak was found in Germany and given the name of *Archaeopteryx* (ancient wing). These creatures lived almost 150 million years ago, and are thought to have been the evolutionary link between dinosaurs and birds.

Louis Pasteur
See pages 142–143

1865

Making food safe

Asked why beer went sour, French chemist Louis Pasteur found it was due to harmful microbes (tiny organisms) in the air. However, the microbes were destroyed if the beer was heated briefly. Wine and milk could be treated in the same way. The heating process, still used today, is called pasteurization.



Milk containers

Heating apparatus

Water bath

19th-century machine for pasteurizing milk

1865



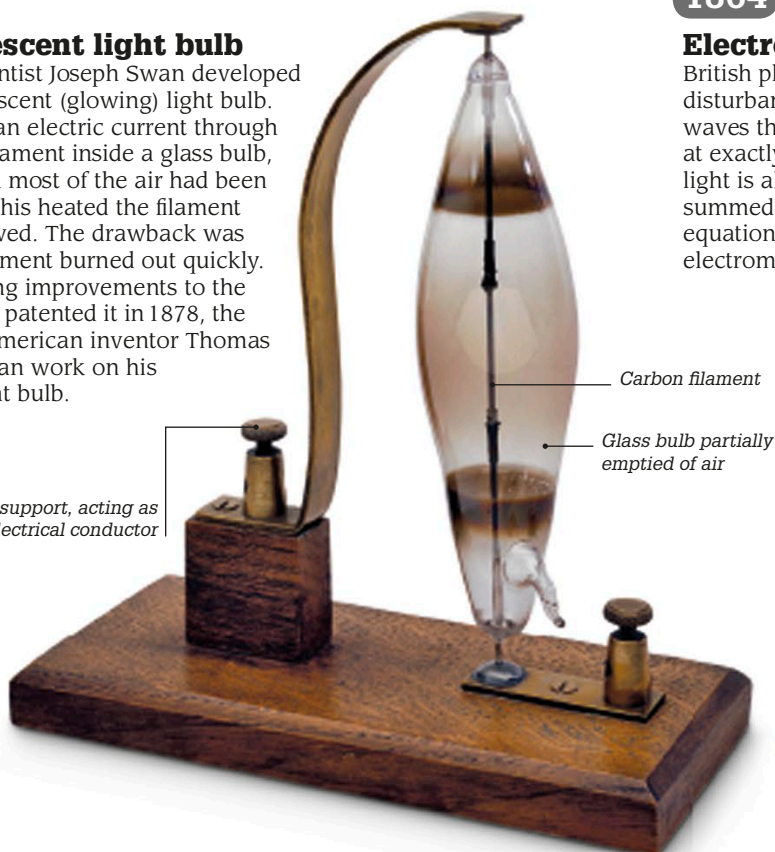
1860

Incandescent light bulb

British scientist Joseph Swan developed an incandescent (glowing) light bulb. He passed an electric current through a carbon filament inside a glass bulb, from which most of the air had been removed. This heated the filament until it glowed. The drawback was that the filament burned out quickly. After making improvements to the bulb, Swan patented it in 1878, the year that American inventor Thomas Edison began work on his electric light bulb.

Brass support, acting as an electrical conductor

Replica of Swan's lamp



1864

Electromagnetism explained

British physicist James Clerk Maxwell showed that disturbances in electromagnetic fields create waves that radiate outward. These waves radiate at exactly the same speed as light, proving that light is also an electromagnetic wave. Maxwell summed up his findings in four mathematical equations that laid the foundations of electromagnetic field theory.



In 1860, Florence Nightingale established the first school of nursing at St. Thomas' Hospital, London.

Thomas Edison
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